Liquid Haskell
Haskell as a Theorem Prover

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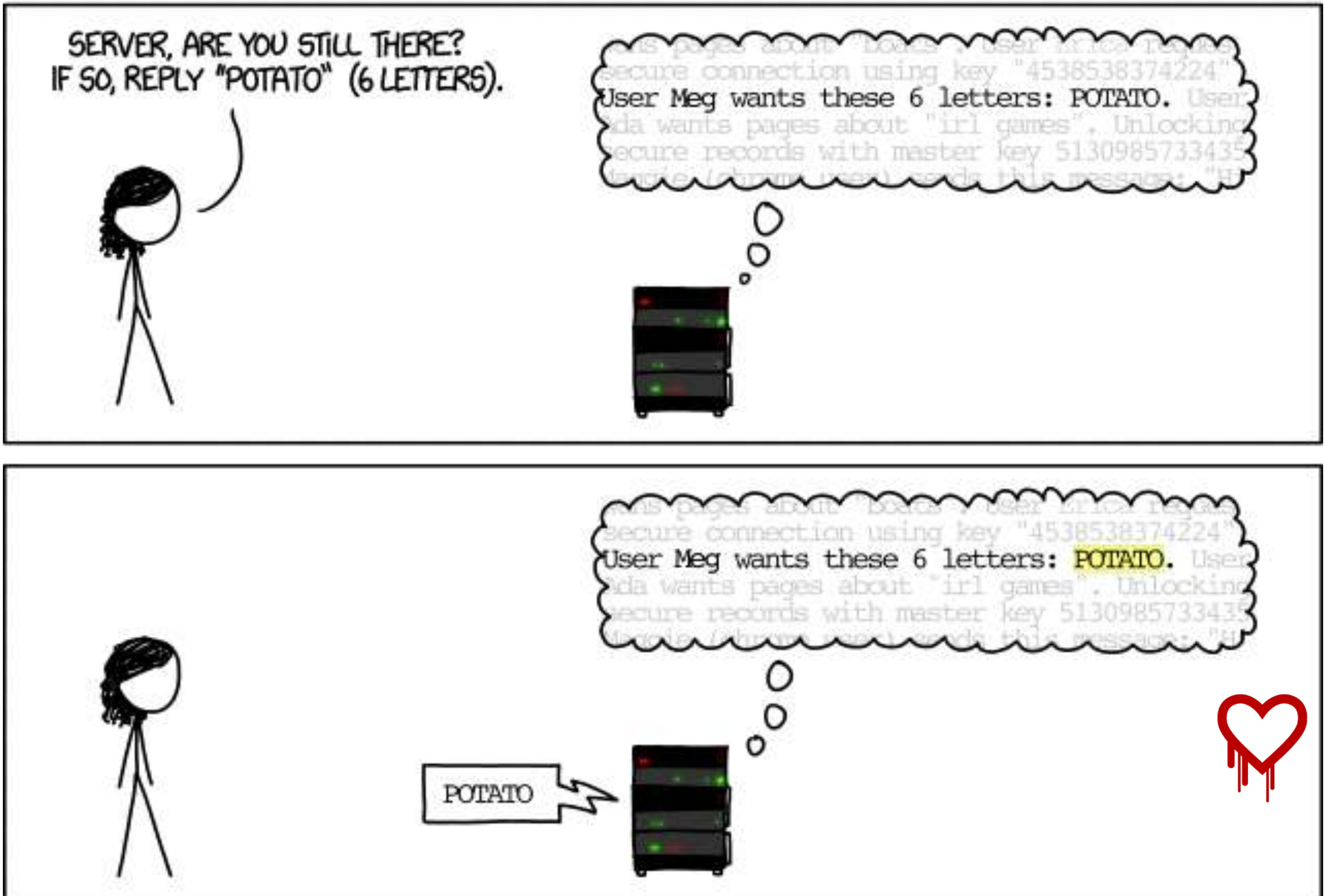
Software bugs are everywhere

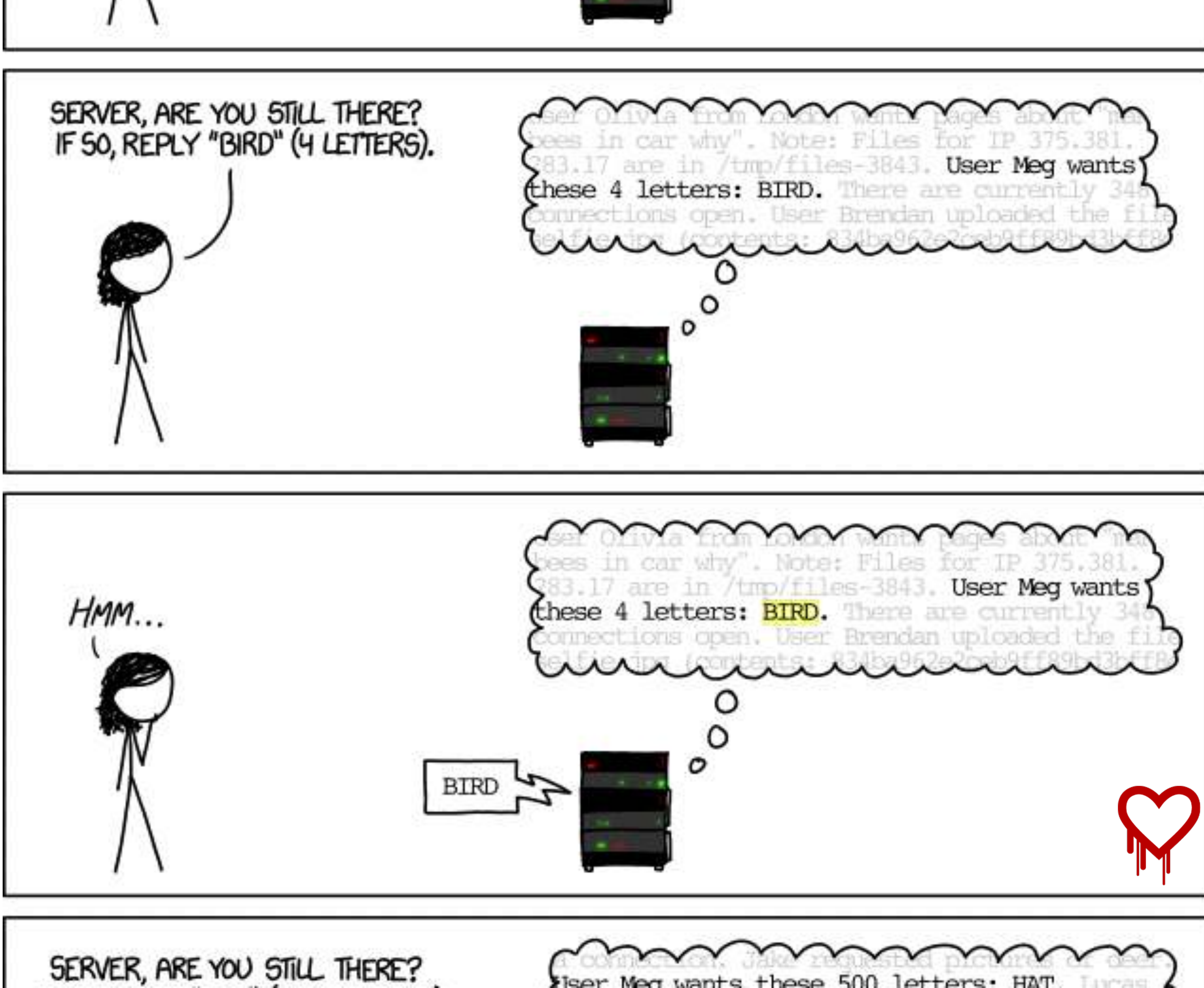
Airbus A400M crashed due to a software bug.
— May 2015

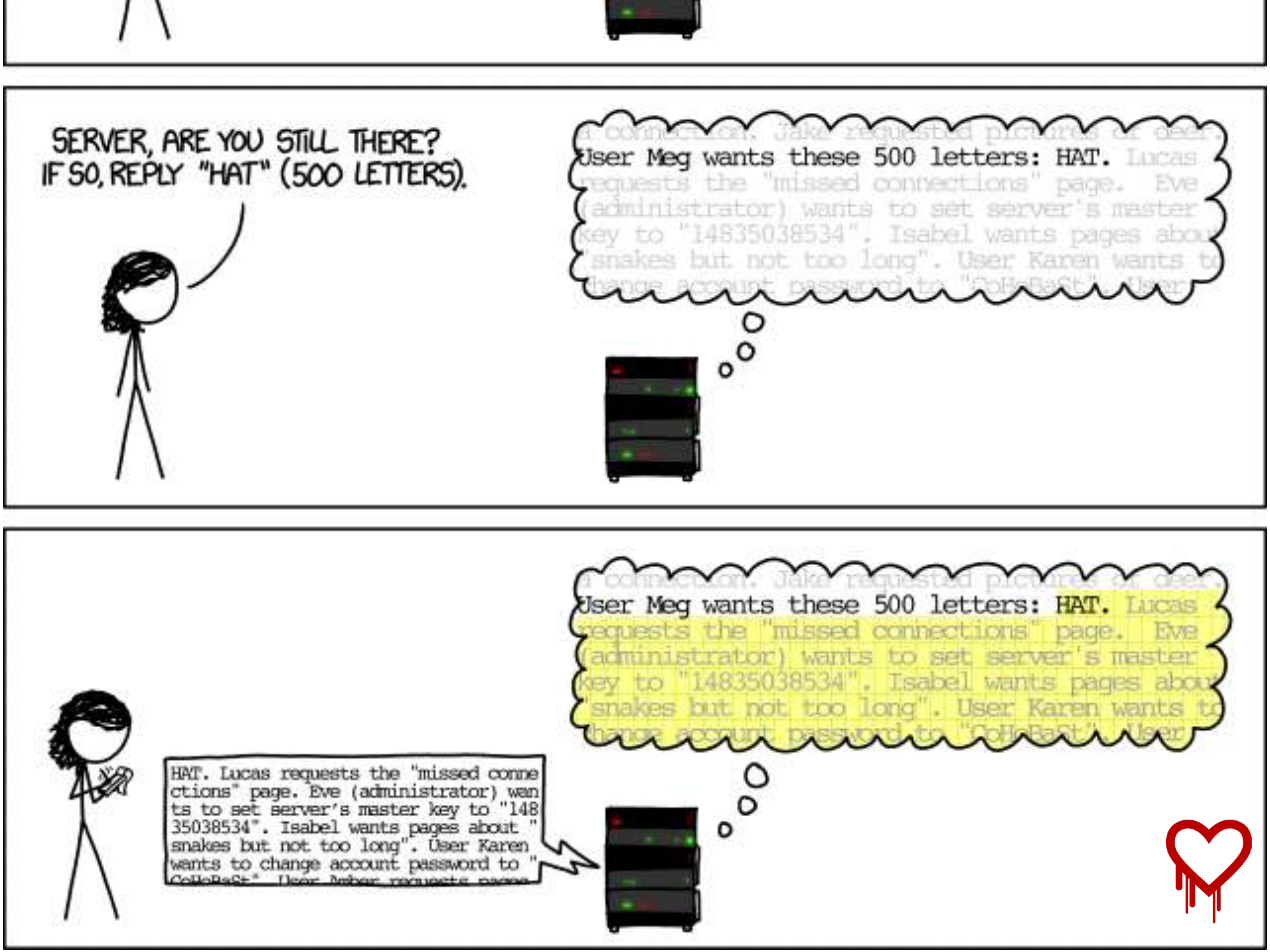
Software bugs are everywhere

The Heartbleed Bug.
Buffer overflow in OpenSSL. 2015

HOW THE HEARTBLEED BUG WORKS:




SERVER, ARE YOU STILL THERE? IF SO, REPLY "BIRD" (4 LETTERS).
User Olivia from London wants pages about "bees in car why". Note: Files for IP 375.381.283.17 are in /tmp/files-3843. User Meg wants these 4 letters: BIRD. There are currently 346 connections open. User Brendan uploaded the file
HMM...
User Olivia from London wants pages about "bees in car why". Note: Files for IP 375.381.283.17 are in /tmp/files-3843. User Meg wants these 4 letters: BIRD. There are currently 346 connections open. User Brendan uploaded the file
BIRD
SERVER, ARE YOU STILL THERE?
User Meg wants these 500 letters: HAT.



SERVER, ARE YOU STILL THERE? IF SO, REPLY "HAT" (500 LETTERS).
a connection. Jake requested pictures of deer.
User Meg wants these 500 letters: HAT. Lucas
requests the "missed connections" page. Eve
(administrator) wants to set server's master
key to "14835038534". Isabel wants pages about
snakes but not too long". User Karen wants to
HAT. Lucas requests the "missed conne
ctions" page. Eve (administrator) wan
ts to set server's master key to "148
35038534". Isabel wants pages about "
snakes but not too long". User Karen
wants to change account password to "

Make bugs difficult to express

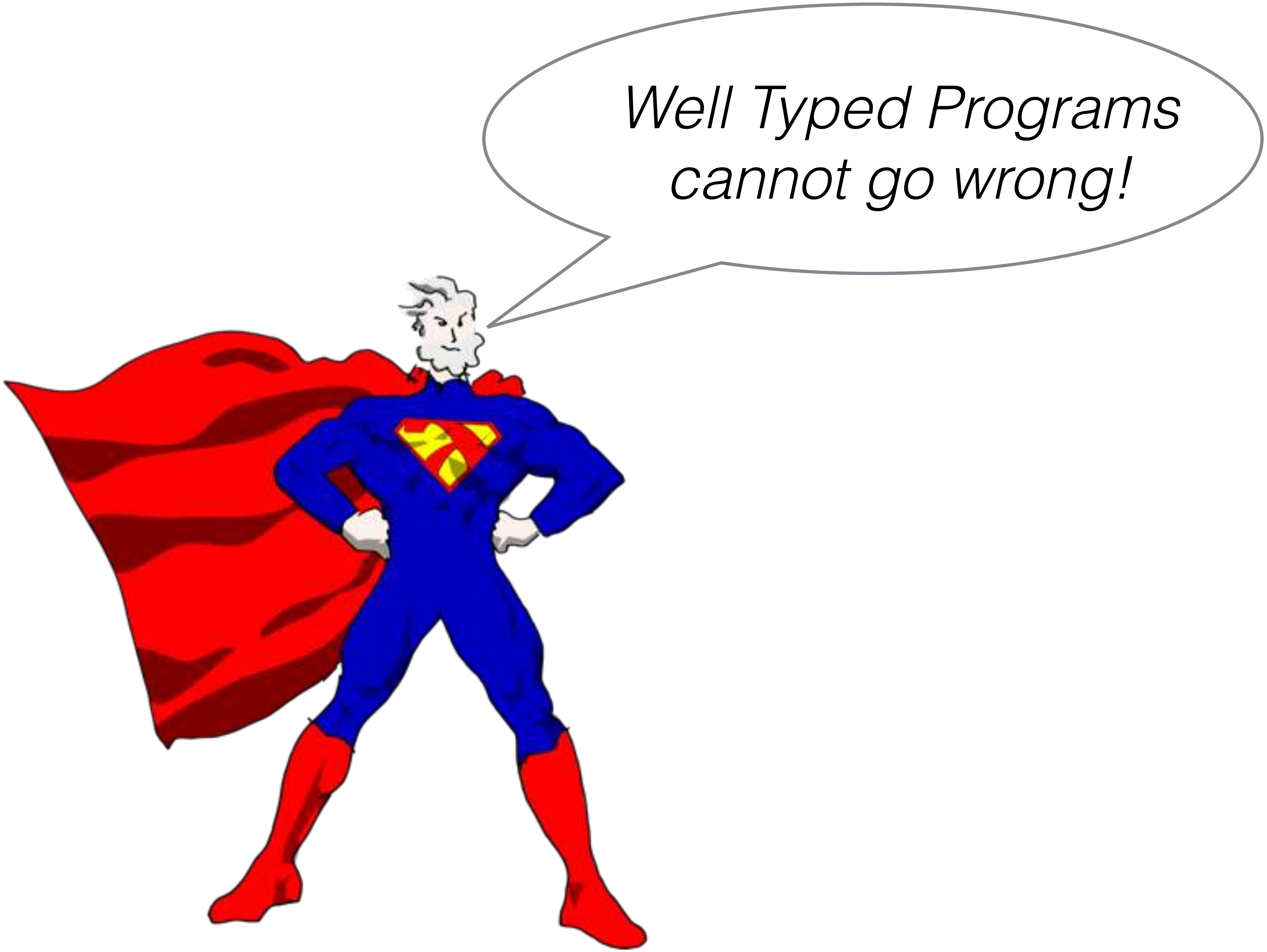
Using Modern Programming Languages

F#, Ocaml, Erlang, Scala, Haskell

Because of

Strong Types + λ-Calculus

Make bugs difficult to express



VS.

VS.

```
λ> :m +Data.Text Data.Text.Unsafe
λ> let pack = "hat"

λ> :t takeWord16
   takeWord16 :: Text -> Int -> Text
```

VS.

```
λ> :m +Data.Text Data.Text.Unsafe
λ> let pack = "hat"

λ> takeWord16 pack True
   Type Error: Cannot match Bool vs Int
```

VS.

```
λ> :m +Data.Text Data.Text.Unsafe
λ> let pack = "hat"

λ> takeWord16 pack 500
   “hat\58456\2594\SOH\NUL…
```

vs.

Valid Values for `takeWord16`?

```
takeWord16 :: t:Text -> i:Int -> Text
```

All Ints

…, -2, -1, 0, 1, 2, 3, …

Valid Values for `takeWord16`?

```
takeWord16 :: t:Text -> i:Int -> Text
```

Valid Ints	Invalid Ints
0, 1 … , len t	len t + 1, …

Refinement Types

```
take :: t:Text -> {v:Int | v <= len t} -> Text
```

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0, 1 … , len t	len t + 1, …

Refinement Types

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take :: t:Text -> {v:Int | v <= len t} -> Text
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λ> let pack = "hat"

λ> take pack 500
   Refinement Type Error
```

Refinement Types

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take :: t:Text -> {v:Int | v <= len t} -> Text
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```
λ> :m +Data.Text Data.Text.Unsafe
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   Refinement Type Error
```

Liquid Haskell

Refinement Types

Checks valid arguments, under facts.

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```
take :: t:Text->{v|v <= len t}->Text

heartbleed = let x = "hat"
             in  take x 500
```

```
len x=3 => v=500 => v<=len x
```

Checks valid arguments, under facts.

```
take :: t:Text->{v|v <= len t}->Text

heartbleed = let x = "hat"
             in  take x 500
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```

Checks valid arguments, under facts.

```
take :: t:Text->{v|v <= len t}->Text

heartbleed = let x = "hat"
             in  take x 500
```

$$\texttt{len x=3} \Rightarrow \boxed{\texttt{v=500}} \Rightarrow \texttt{v<=len x}$$

Checks valid arguments, under facts.

```
take :: t:Text -> {v | v <= len t} -> Text

heartbleed = let x = "hat"
             in  take x 500
```

```
len x = 3 => v = 500 => v <= len x
```

Checks valid arguments, under facts.

```
take :: t:Text->{v|v <= len t}->Text

heartbleed = let x = "hat"
             in  take x 500
```

$\boxed{\texttt{len x=3 => v=500 => v<=len x}}$

Checks valid arguments, under facts.

```
take :: t:Text->{v|v <= len t}->Text

heartbleed = let x = "hat"
             in  take x 500
```

SMT-query

```
len x=3 => v=500 => v<=len x
```

Checks valid arguments, under facts.

```
take :: t:Text->{v|v <= len t}->Text

heartbleed = let x = "hat"
             in  take x 500
```

SMT-Invalid

```
len x=3 => v=500 => v<=len x
```

Checks valid arguments, under facts.

```
take :: t:Text->{v|v <= len t}->Text

heartbleed = let x = "hat"
             in  take x 500
```

Checker reports **Error**

Liquid Haskell:

Checks valid arguments, under facts.

What are interesting facts?

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```
len “hat” = 3
```

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(xs ++ ys) ++ zs == xs ++ (ys ++ zs)
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f is == mapReduce f is
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Theorems about Haskell functions

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(Liquid) Haskell as a theorem prover.

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Specify theorems as Refinement Types

Prove theorems in Haskell

Use Liquid Haskell to check correctness

(Liquid) Haskell as a theorem prover.

```
data L a = N | C a (L a)
```

(Liquid) Haskell as a theorem prover.

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```
N ++ ys          = ys
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(Liquid) Haskell as a theorem prover.

```
N ++ ys           = ys
(C x xs) ++ ys = C x (xs ++ ys)
```

```
assoc :: xs:L a -> ys:L a -> zs:L a ->
             ()
```

(Liquid) Haskell as a theorem prover.

```
N ++ ys          = ys
(C x xs) ++ ys = C x (xs ++ ys)
```

```
assoc :: xs:L a -> ys:L a -> zs:L a ->
  {v:()|(xs ++ ys)++zs==xs++(ys++zs)}
```

(Liquid) Haskell as a theorem prover.

```
N ++ ys          = ys
(C x xs) ++ ys = C x (xs ++ ys)
```

```
assoc :: xs:L a -> ys:L a -> zs:L a ->
 { (xs ++ ys) ++ zs == xs ++ (ys ++ zs) }
```

(Liquid) Haskell as a theorem prover.

What is the body of assoc?

```
assoc :: xs:L a -> ys:L a -> zs:L a ->
 { (xs ++ ys) ++ zs == xs ++ (ys ++ zs) }
assoc xs ys zs = ???
```

(Liquid) Haskell as a theorem prover.

What is the body of assoc?

A unit Haskell value showing that left-hand side == right-hand side

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assoc :: xs:L a -> ys:L a -> zs:L a ->
 { (xs ++ ys) ++ zs == xs ++ (ys ++ zs) }
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(Liquid) Haskell as a theorem prover.

What is the body of assoc?

A unit Haskell value showing that
left-hand side == right-hand side

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assoc :: xs:L a -> ys:L a -> zs:L a ->
  { (xs ++ ys) ++ zs == xs ++ (ys ++ zs) }
assoc xs ys zs = ()
```

Type Error!

(Liquid) Haskell as a theorem prover.

A unit Haskell value showing that
left-hand side == right-hand side

```
assoc :: xs:L a -> ys:L a -> zs:L a ->
 { (xs ++ ys) ++ zs == xs ++ (ys ++ zs) }
assoc xs ys zs
 =   (xs ++ ys) ++ zs
     …
 ==. xs ++ (ys ++ zs)
 *** QED
```

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assoc :: xs:L a -> ys:L a -> zs:L a ->
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assoc :: xs:L a -> ys:L a -> zs:L a ->
 { (xs ++ ys) ++ zs == xs ++ (ys ++ zs) }
assoc N  ys zs
 =   (N  ++ ys) ++ zs


 ==. N  ++ (ys ++ zs)
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 ==. ys          ++ zs
 ==. N  ++ (ys ++ zs)
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Demo

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Thanks!

END